

JIAPING LIU

PR in Canada | jiapivialiu@gmail.com | [GitHub: jiapivialiu](#) | [LinkedIn: jiaping9](#) | Web: jiapivialiu.com

EDUCATION

-
- | | |
|--|-------------|
| University of British Columbia — PhD in Statistics | 2020 - 2026 |
| - Research Area : Computational Statistics, Convex Optimization, Machine Learning Algorithm | |
| University of Ottawa — MSc in Statistics (GPA : 3.9/4.0) | 2018 - 2020 |
| Zhejiang Univ. of Finance & Economics — BA in Economic Statistics (GPA : 3.7/4.0) | 2014 - 2018 |

WORK EXPERIENCE

-
- | | |
|---|---------------------|
| Machine Learning Engineer (Recommendation Systems), JD.com | Dec 2025 - Mar 2026 |
| - Improved the click-through rate (CTR) by better ranking the retrieved products in the push/messaging product recommendation scenario (using feature engineering and importance analysis, deep learning (including deep cross models, factorization machines, multi-head target attention, etc.), joint pointwise & pairwise learning, and large language models, etc.) with +0.31% relative overall pCTR increase and +0.78% among high-active customers. | |
| - Built SQL-based monitoring panels for online/offline metrics tracking; conducted online A/B experimentation to validate model iterations; explored natural language-based query understanding for personalized recommendation; addressed engineering problems including online-offline inconsistency and cold start in low-active users. | |
| - <i>Toolkit: Python, TensorFlow, SQL, Spark & Hive/Hadoop, Git, HPC, LLM.</i> | |
| Risk Modeling Analyst (Credit Risk & Machine Learning), Royal Bank of Canada | Sep - Dec 2025 |
| - Built a credit risk modeling pipeline on billion-level data given user behavior features using survival analysis as a baseline to predict the probability of default on the overdraft accounts in RBC. | |
| - Addressed complex and large-scale data problems including duplicated records, large proportion of data missing, data mistakes in account open/reopen/close dates, which much improved data quality. | |
| - Explored LLM-derived transaction embeddings to exploit high-order user behavior; reimplemented survival-model fitting in PyTorch, cutting training time by >4× on million-row data with distributed scaling support. | |
| - <i>Toolkit: Python, PySpark, AWS/S3, SQL, PyTorch, Scala, Git, HPC, LLM.</i> | |
| Statistical Consultant, Applied Statistics and Data Science Group, UBC | May 2021 - Aug 2023 |
| - Collaborated with 10+ clients across diverse research domains, applying machine learning and statistical modeling; independently authored analysis reports with data-driven solutions. | |

RESEARCH PROJECTS

-
- | | |
|--|---------------------|
| Nonparametric Time Series Analysis via Trend Filtering and Convex Optimization | Sep 2021 - Aug 2026 |
| <i>Two first & corresponding author papers, two ongoing projects.</i> | |
| - <i>Paper 1</i> (Published in <i>PLOS Computational Biology</i> , 2024) : Proposed an ℓ_1 trend filtering model for estimating time-varying reproduction numbers of infectious diseases, achieving 10%–30% accuracy improvement (measured by KL divergence) over existing methods. Implemented core optimization algorithms (ADMM, proximal Newton, dynamic programming) in C++. | |
| - <i>Paper 2</i> (Under review) : Developed a linear-time algorithm based on <i>state-space models and Kalman filter</i> to accurately solve ill-conditioned linear systems structured on discrete derivatives, resolving numerical instability of existing methods under extreme non-uniform spacing. Completed all mathematical derivations and C++ implementation. | |
| - <i>Toolkit: R, C/C++, Python, Quarto, $\LaTeX$, GitHub/Git, HPC. Developed: rtestim, glmgen/trendfilter.</i> | |

HACKATHON PROJECTS

-
- | |
|---|
| Vela AI : Multimodal Localization for E-Commerce — GMI Cloud Hackathon 2026 |
| - Built a <i>multi-agent</i> VLM/LLM pipeline for cross-border ecommerce (China $\rightarrow$ Canada), automating product image understanding, bilingual listing generation, and marketing photo creation with end-to-end LLM Agent content review. |
| AI-Generated Text Detection (SOTA) — Microsoft AI Hackathon 2024 |
| - <i>SFT</i> a lightweight LLM via LoRA/PEFT for AI-generated text detection; achieved 93.9% accuracy (SOTA) on RAID benchmark and 99.3% on GPT-generated text. Open-sourced on HuggingFace. |